

ECON 1550

Spring 2026

Problem Set 9

Due: April 24, 2026 at 11:59pm ET

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Submission: Canvas or Gradescope

Instructions

- When submitting to Gradescope, indicate the page where each question is answered to avoid a 5-point deduction.
- Full credit is given for correct answers. If multiple steps are needed, you must show them to get full credit.
- Points are shown for each part. Partial credit is given for partially correct answers; show your work to maximize it.
- Late submissions receive a score of zero.
- If you have technical problems submitting, email your work to the Head TA before the deadline.
- Collaboration with classmates is encouraged; use of generative AI is permitted but discouraged.
- You must write, understand, and submit your solutions individually. Copying other students' or AI-generated answers, even fragments, is not allowed.

1. Devaluation and the Current Account (12 points)

Assume the economy is at its long-run equilibrium and that the current account is in balance (equal to zero). The central bank is implementing a fully credible fixed exchange rate regime. Using the AA-DD model, explain why a devaluation improves the current account in the short run.

2. A Flat AA Curve (32 points)

Consider the following version of the AA-DD model:

Consumption : $C = (1 - a)Y$

Investment : I

Government spending : G

Current account	:	$CA = b \frac{EP^*}{P} - cY$
Demand for domestic goods	:	$D = C + I + G + CA$
Real money demand	:	$\frac{M^d}{P} = 1 + d(Y - Y^f) - eR$
Real money supply	:	$\frac{M^s}{P}$
Uncovered interest parity	:	$R = R^* + \frac{E^e}{E} - 1$

The variables P^* , I , G , Y^f , M^s , and R^* are exogenous. The parameters a , b , c , d , and e are also exogenous and assumed to be positive. The variables C , Y , CA , E , P , D , E^e , M^d , and R are endogenous.

Assume that:

- in the long run, $Y = Y^f$, $E = E^e$ and P is fully flexible;
 - in the short run, P is fixed, and E^e is equal to the value of E in the final long-run equilibrium;
 - the economy is initially in a long-run equilibrium with output equal to the full-employment level Y^f .
- (a) [8 points] Give intuition for why the real money demand is increasing in Y and identify the parameter that determines how real money demand responds to Y .
 - (b) [8 points] Show in an AA-DD diagram the effects of a permanent change in the parameter d from its initial positive value to a new value of $d = 0$, and state whether the equilibrium (Y, E) changes.
 - (c) [8 points] For this question, assume that the economy starts at the long-run equilibrium that is reached after d changes to zero permanently, and continue to assume that $d = 0$ permanently. Consider a temporary fiscal expansion that increases G to a higher value G^{high} . In the short run, does the current account increase, decrease, or stay the same? Give economic intuition for your answer.
 - (d) [8 points] Now assume that, before the temporary fiscal expansion, the central bank implements a fixed exchange rate regime at the prevailing exchange rate, which is the long-run level with $d = 0$. Are the short-run effects of fiscal expansion on the current account the same as in the last question? Why or why not?

3. A Preview of the II-XX Model (56 points)

This question introduces the II-XX model, a framework for analyzing internal and external balance in an open economy. Everything you need is given below; you do not need prior knowledge of this model. For optional background, you may consult the [lecture notes chapter on the II-XX model](#), but it is not required to answer any part of this question.

The current account is given by:

$$CA(q, Y) = d_1 q - d_2 Y, \quad (1)$$

where q is the real exchange rate, Y is output, and d_1, d_2 are positive parameters. Domestic demand (absorption) is

$$A = C + G, \quad (2)$$

where G is government spending, and C is consumption, which is given by

$$C = c_0 + c_1 Y, \quad (3)$$

where c_0 is a positive parameter and $c_1 \in (0, 1)$ is the marginal propensity to consume. We denote the full-employment level of output by Y^f and treat it as exogenous.

The II curve in this economy is given by pairs A, q that make the goods-market be in equilibrium when output is at full employment:

$$Y^f = A + CA(q, Y^f). \quad (4)$$

The XX curve is given by pairs A, q that keep the current account constant at its long-run sustainable level XX^b :

$$XX^b = CA(q, A), \quad (5)$$

where XX^b is an exogenous constant level for the current account, and $CA(q, A) = CA(q, Y)$ is the current account expressed as a function of q and A instead of q and Y .

The economy is in “internal balance” when $Y = Y^f$ and in “external balance” when $CA = XX^b$. Both are stable rest points: absent shocks, Y converges to Y^f and CA converges to XX^b , and both remain there once reached.

- (a) [8 points] Assume that, for imports, the volume effect dominates the value effect. Give economic intuition for why CA is increasing in q and decreasing in Y .
- (b) [8 points] Use equations (2) and (3) to express Y as a function of A . Then find the function $CA(q, A)$ explicitly.

- (c) [8 points] Find the function $f(\cdot)$ such that equation (4) can be written as $q = f(A)$. Is f increasing, decreasing, or constant? Give economic intuition.
- (d) [8 points] Find the function $g(\cdot)$ such that equation (5) can be written as $q = g(A)$. Is g increasing, decreasing, or constant? Give economic intuition.
- (e) [8 points] Sketch the Π and XX^b curves in a diagram with A on the horizontal axis and q on the vertical axis. Label the intersection as (A^*, q^*) .
- (f) [8 points] Assume the economy is initially at the intersection of the Π and XX^b curves. The government permanently increases G . Do the Π and XX^b curves shift? Explain why, using your expression for $CA(q, A)$ from (b).
- (g) [8 points] Immediately after the increase in G , assume q and Y are unchanged on impact, so $q = q^*$ and $Y = Y^f$. What is the new value of A ? Is the economy in internal balance, external balance, both, or neither? Explain.